Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/27/2025
PI Name:	Bowles, Dawn Ph.D.		
Project Title:	A Multi-Omics and Multi-Species Examination of Combined Environmental Stressors of Space Exploration		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dawn.bowles@duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919-668-1947
Organization Name:	Duke University		
PI Address 1:	Department of Surgery		
PI Address 2:	Msrb1 Room 401B, DUMC 2642		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27710-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	12/01/2021	**End Date:**	11/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 12/31/24).		
Key Personnel Changes/Previous PI:	We have had two changes in personnel since the beginning of the grant. First Franklin Lee was the original technician on the grant. He was accepted into medical school for the Fall of 2022. Matthew Smith was trained on the position and took Franklin's place once Franklin departed for medical school. Matthew Smith departed on April 1 2023 for medical school. Alejandro Lobo, MD was hired and trained to replace Matthew Smith.		
COI Name (Institution):	Kidane, Yared Ph.D. (Self)		
Grant/Contract No.:	80NSSC22K0253		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The dominant environmental stressors that may affect cardiac biology during space exploration are space radiation and weightlessness. The key central objective of this proposal is to obtain a multi-omics and physiological understanding of the impact of space radiation and weightlessness individually and together on the cardiovascular system using two rodent models: rat neonatal cardiomyocytes (RNNC) and c57BL/6 mice. In studies evaluating the effect of microgravity on the protein content of RNNC, we were able to demonstrate that protein turnover diminishes globally in these cells. Moreover, we were able to pinpoint molecular pathways potentially responsible and possibly targetable for this effect. We have an ongoing study evaluating the cardiac phenome and cardiac proteome in c57BL/6 mice exposed to galactic cosmic rays (GCR). Our data suggests that, in contrast to other forms of radiation, a single exposure to GCR results in a significant deterioration of cardiac function that becomes apparent long after exposure (9-12 months post GCR). Bioinformatics analyses of the cardiac proteome of these mice have identified that keratinization and cornification pathways are up regulated in hearts from GCR irradiated mice. These pathways are presumably upregulated as a response to sustained inflammation and persistent oxidative damage post radiation. The impact of both stressors simultaneously (GCR and weightlessness) on cardiac physiology and cardiac proteomics has not been explored at all. We will leverage our expertise using these models from two species (mouse and rat), as well as accumulated data sets and biospecimens accrued from other studies, to address the following two aims: Aim 1. Test the hypothesis that the combination of GCR exposure and weightlessness will result in additive, or even synergistic, enhancements in cell death, increases in reactive oxygen species, protein turnover, and mitochondrial dysfunction over that observed with either stressor alone. This aim will utilize the RNNC model and the Synthecon Rotary Cell Culture System for weightlessness. GCR, with and without weightlessness, will be administered to RNNC at the NASA Space Radiation Laboratory (NSRL). Functional readouts will include assessments of cell viability, cell death (apoptosis), and mitochondrial integrity and functionality, as well as multi-omics comparative assessments (transcriptomics and mass spectrometry based proteomics). Aim 2. Test the hypothesis that the cardiac dysfunction previously observed in c57BL/6 at 9-12 months post GCR treatment will manifest with a quicker onset and more pronounced decrements in functional measurements in c57BL/6 undergoing both GCR irradiation and hind limb unloading (model of weightlessness). Other readouts will include assessment of cardiac tissue histology as well as multi-omics comparative assessments (transcriptomics and mass spectrometry based proteomics). Upon completion of this study, we will have generated multiple proteomics data sets as deliverables that will be bioinformatically mineable by us and by others. This data will enable a greater understanding of the organism's response to space radiation and weightlessness environments alone and in combination. This proposal is relevant to both the Cell and Molecular Biology Element and the Animal Biology (Vertebrate). This proposal aligns to many of the guiding questions of these elements such as: CMB-1 What are the underlying genetic, molecular, and biochemical mechanisms of the cell that are influenced by gravitational changes and the space environment? CMB-2 What are the systems biology mechanisms and pathways for cell function that are responsive to gravitational changes and the spaceflight conditions? CMB-3 What are the cellular and molecular bases for gravitational changes and space environment effects to tissue, organ, and whole organisms? CMB-5 Does the space flight environment affect or influence cell and molecular functions causing tissue/organ dysfunction or disease states?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This grant is exploring whether pathways perturbed in the heart or heart cells by the combined environmental stressors of microgravity and space radiation are similarly perturbed in two animal species (rats and mice). Should common pathway be elucidated from this study, this may lead to interventions that might alleviate fragility in patients with end stage heart failure or patients that are mainly bedridden or immobile. There are no innovative technologies to report at this point of the study.
Task Progress:	Aim 1. Experiments in Aim 1 were performed at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). Rat neonatal cardiomyocytes (RNNCs) were isolated from 2-day old rat neonates and cultured in stable isotope labeling by amino acids in cell culture (SILAC) media (Fisher Scientific) for a period of 120 hours prior to irradiation. Cells were divided in 4 experimental groups: A) Microgravity and Irradiation B) Microgravity alone C) Radiation alone D) Control group (no stress). Cells that were intended to be exposed to microgravity were cultured in the Rotary Cell Culture System TM (RCCS, Synthecon, Inc.), and the remainder of the cells were cultured in regular T-25 flasks (Corning). This experiment was done 4 different times in a period of 6 weeks at BNL. All the experimental cells were transported the day of irradiation and exposed to the same external stressors. Cells that were irradiated were placed in front of the beam, specifically the 20x20cm sized beam, and a received a simple Galactic Cosmic Ray simulation (GCR sim) at 75cGy dose. 24 hours post irradiation cells were harvested. Our first samples of cell pellets were sent to the Duke Proteomics and Metabolomics Core Facility in the winter of 2023-2024. Samples were processed and subjected to mass spectrometry proteomics pipeline, analyzed, and a final report was generated by Dr. Matthew Foster. For quality control purposes, they assessed data in Spectronaut using coefficient of variation (CV) plots and principal component analysis. Reproducibility/precision was high based on % CVs of the replicate SPQC samples (median 5.3%) and clustering by principal component analysis (PCA), and there was good intra-group reproducibility (median % CV of ~15%). Differential protein abundance was determined in Spectronaut using a paired t-test, with or without imputation of missing data with background signal. For the latter, data was filtered on precursors that were identified in >50% of runs, with background signal imputation. There were 110,153 precursors and 7,682 proteins that met these filtering criteria. Using the unlabeled workflow, they were able to quantify >8,000 protein groups (without imputation) and almost 7,700 protein groups after filtering of precursors that were quantified in >50% of samples. There was a striking difference between cells cultured at gravity versus microgravity.

	Our analysis continued in the early spring of 2024, with the collaboration of Dr. Yared Kidane, an expert in bioinformatics. Dr. Kidane started his analysis generating violin plots for normalized data and graphing empirical distribution of the data set for the different experimental groups and different replicates of the experiment. Dr. Kidane also did a principal component analysis, showing an overwhelming difference amongst the regular gravity (RG) and microgravity (MG) groups, confirming what was seen by Dr. Foster. Hierarchical clustering depicted a clear grouping difference between regular gravity and microgravity, resonating with the principal component analysis. Posterior analysis included mean difference plots and volcano plots comparing the different experimental groups. Mean difference plots showed the average intensity of down regulated and upregulated differentially expressed proteins in microgravity (MG) versus regular gravity (RG) in no radiation environment and in microgravity (MG) vs regular gravity (RG) in radiation environment. Volcano plots were created, representing the overall perturbation significance of proteins based on p-values and fold-change criteria, in MG vs RG in no radiation and in MG vs RG in radiation . The purpose of this analysis was to understand the proteomic alterations and the number of differentially expressed proteins in the different comparisons. Posterior analysis focused in trying to isolate the effects of radiation and microgravity and then evaluate them synergistically; this was done for both the proteins individually and biological pathways. Different notations were created for the experimental groups, for abbreviation purposes. Venn diagrams comparing experimental groups were generated; upregulated and downregulated differentially expressed proteins were quantified. In the Space (Synergistic effect of stressors) vs. Earth comparison, there are 37 proteins identified as downregulated and 48 proteins identified as upregulated. Volcano plots were generated to understand the proteomic alterations generated by microgravity, radiation, and the synergism of stressors. At this point, it was evident the overriding effect generated by microgravity alone in comparison to radiation alone, observed in the quantity of differentially expressed proteins. Assessments on pathways from KEGG, REACTOME, and GO biological processes revealed significant perturbation in pathways in each data set. We are currently analyzing the biology of the pathway perturbation analysis and planning validation experiments, in our weekly meetings, to confirm the reports from bioinformatics.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Kidane YH, Lee FH, Smith MF, Wang C, Mirza JB, Sharma S, Lobo AA, Dewan KC, Chen J, Diaz TE, Pla MM, Foster MW, Bowles DE. "Proteomic and phosphoproteomic characterization of cardiovascular tissues after long term exposure to simulated space radiation." Front Physiol. 2024 Apr 18;15:1248276. https://doi.org/10.3389/fphys.2024.1248276 ; PMID: 38699144; PMCID: PMC11063234 , Apr-2024